

ORF309 Midterm 1 - Clear Cheat Sheet

1 Conditional Probability (Key for Every Exam!)

1.1 The Basic Formula

Definition: Probability of A happening, given we know B happened:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

What this means: We "zoom in" to the world where B happened, then ask what fraction of that world contains A .

Rearranged (multiplication rule):

$$P(A \cap B) = P(A|B) \cdot P(B)$$

1.2 Chain Rule (Multiple Events)

For several events happening together:

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2)$$

Process: Start with first event, then condition each next event on all previous ones.

2 Law of Total Probability (LOTP)

2.1 When to Use It

Use LOTP when you want to find $P(A)$ but the path to A goes through different "cases" or "stages".

2.2 The Formula

If $B_1, B_2, \dots, B_n$ partition the sample space (disjoint and cover everything):

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$$

Two-case version (most common):

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

2.3 Example: Defective Widgets

Factory has Line A (produces 60% of widgets, 2% defective) and Line B (produces 40%, 5% defective). What's probability a random widget is defective?

Solution: Let D = defective, A = from Line A, B = from Line B

$$\begin{aligned} P(D) &= P(D|A)P(A) + P(D|B)P(B) \\ &= 0.02 \times 0.6 + 0.05 \times 0.4 = 0.012 + 0.020 = 0.032 \end{aligned}$$

3 Bayes' Theorem (Updating Beliefs)

3.1 The Formula

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$$

In words:

- $P(B_j)$ = prior (what we thought before seeing data A)
- $P(A|B_j)$ = likelihood (how likely is data A if B_j is true)
- $P(B_j|A)$ = posterior (updated belief after seeing data A)

3.2 Example: Defective Widget Origin

Same factory. Given a widget is defective, what's probability it came from Line A?

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)} = \frac{0.02 \times 0.6}{0.032} = \frac{0.012}{0.032} = 0.375$$

Interpretation: Even though Line A produces 60% of widgets, only 37.5% of defective widgets come from it (because Line B has higher defect rate).

4 Independence

4.1 What Independence Means

A and B are independent if knowing one tells you nothing about the other:

$$P(A \cap B) = P(A) \cdot P(B)$$

Equivalently: $P(A|B) = P(A)$ (when $P(B) > 0$)

4.2 Critical Warning!

Mutually Exclusive $\neq$ Independent!

If $A \cap B = \emptyset$ (disjoint) and $P(A), P(B) > 0$:

- Then A and B are **DEPENDENT**
 - Why? If A happens, you know for sure B didn't happen
- Example:** Rolling a die. Let A = "roll 1 or 2", B = "roll 3 or 4". These are disjoint but dependent (knowing one happened tells you the other didn't).

4.3 Properties of Independence

If $A \perp B$ (independent), then also:

- $A \perp B^c$ (independent of complement)
- $A^c \perp B$
- $A^c \perp B^c$

5 Discrete Distributions (Know These Cold!)

5.1 Geometric Distribution: Geom(p)

Models: Number of trials until first success (when each trial succeeds with probability p)

PMF: $P(X = k) = (1 - p)^{k-1} \cdot p$ for $k = 1, 2, 3, \dots$

Key formulas:

- $E[X] = \frac{1}{p}$ (average trials until success)
- $P(X > k) = (1 - p)^k$ (haven't succeeded yet after k tries)
- $\text{Var}(X) = \frac{1-p}{p^2}$

MEMORYLESS PROPERTY (unique to Geometric!):

$$P(X > m + n | X > m) = P(X > n)$$

What this means: If you've already waited m trials without success, the probability of waiting at least n more is the same as if you were starting fresh.

Example: Bus comes with probability 0.1 each minute. You've waited 10 minutes. What's probability you wait at least 20 more?

$$P(X > 30 | X > 10) = P(X > 20) = (0.9)^{20}$$

5.2 Binomial Distribution: Bin(n, p)

Models: Number of successes in n independent trials (each succeeds with probability p)

PMF: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$ for $k = 0, 1, \dots, n$

Key formulas:

- $E[X] = np$
- $\text{Var}(X) = np(1 - p)$

Example: Flip fair coin 10 times. Number of heads is $\text{Bin}(10, 0.5)$:

- $E[\text{heads}] = 10 \times 0.5 = 5$
- $P(\text{exactly 7 heads}) = \binom{10}{7} (0.5)^7 (0.5)^3 = \binom{10}{7} (0.5)^{10}$

5.3 Poisson Distribution: Pois(λ)

Models: Number of events in fixed time/space when events happen at rate λ

PMF: $P(N = k) = e^{-\lambda} \frac{\lambda^k}{k!}$ for $k = 0, 1, 2, \dots$

Key formulas:

- $E[N] = \lambda$
- $\text{Var}(N) = \lambda$

When to use: "Rare events" - when n is large, p is small, and $np = \lambda$ is moderate:

$$\text{Bin}(n, p) \approx \text{Pois}(np)$$

6 Expectation (The Average)

6.1 Basic Definition

For discrete random variable X :

$$E[X] = \sum_{\text{all } x} x \cdot P(X = x)$$

What this means: Weighted average of all possible values, weighted by their probabilities.

6.2 Linearity of Expectation (THE MOST USEFUL PROPERTY!)

For any random variables X and Y (even dependent!):

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

Why this is amazing: No independence needed! Works always!

Example: Roll two dice. Expected sum?

Let X_1 = first die, X_2 = second die. Even though they're not independent of their sum, we can use linearity:

$$E[X_1 + X_2] = E[X_1] + E[X_2] = 3.5 + 3.5 = 7$$

6.3 Indicator Trick (Simplify Everything!)

Key idea: Define indicator random variables for events:

$$1_A = \begin{cases} 1 & \text{if event } A \text{ happens} \\ 0 & \text{otherwise} \end{cases}$$

Magic property: $E[1_A] = P(A)$

How to use it:

1. Write X as sum of indicators: $X = \sum_i 1_{A_i}$
2. Use linearity: $E[X] = \sum_i E[1_{A_i}] = \sum_i P(A_i)$

Example: Bob's exam has 2 questions. He gets question 1 right with prob 0.7, question 2 right with prob 0.6. Each worth 10 points. Expected score?

Let $X = \text{score} = 10 \cdot 1_{Q1} + 10 \cdot 1_{Q2}$

$$E[X] = 10 \cdot P(Q1) + 10 \cdot P(Q2) = 10(0.7) + 10(0.6) = 13$$

Note: Works even if questions aren't independent!

6.4 Conditional Expectation

Definition: Expected value of X given event B happened:

$$E[X|B] = \sum_{\text{all } x} x \cdot P(X = x|B)$$

Law of Total Expectation (Tower Property):

$$E[X] = \sum_i E[X|B_i]P(B_i)$$

where $B_1, \dots, B_n$ partition the sample space.

When to use: When computing $E[X]$ directly is hard, but $E[X|\text{something}]$ is easier.

Example: Expected widget defects. If from Line A: $E[\text{defects}|A] = 0.02$, if from Line B: $E[\text{defects}|B] = 0.05$. Line A makes 60% of widgets.

$$\begin{aligned}
 E[\text{defects}] &= E[\text{defects}|A] \cdot P(A) + E[\text{defects}|B] \cdot P(B) \\
 &= 0.02 \times 0.6 + 0.05 \times 0.4 = 0.032
 \end{aligned}$$

7 Variance (Measuring Spread)

7.1 Definition & Key Formula

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

Second formula is usually easier! Calculate $E[X^2]$ and $E[X]$, then:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

7.2 Properties

- $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (adding constant doesn't change spread)
- $\text{Var}(X) \geq 0$ always
- $\text{Var}(X) = 0$ only if X is constant

7.3 Variance of Sum (NEEDS Independence!)

If X and Y are independent:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Critical: Unlike expectation, this **requires independence!**

Example: Bob's exam (continued). Variance of score?

Assuming questions independent:

$$\begin{aligned}
 \text{Var}(X) &= \text{Var}(10 \cdot 1_{Q1}) + \text{Var}(10 \cdot 1_{Q2}) \\
 &= 100 \cdot \text{Var}(1_{Q1}) + 100 \cdot \text{Var}(1_{Q2})
 \end{aligned}$$

For Bernoulli with p : $\text{Var} = p(1 - p)$

$$= 100[0.7(0.3)] + 100[0.6(0.4)] = 21 + 24 = 45$$

8 Problem-Solving Strategies

8.1 When to Use Each Technique

Use LOTP when:

- Problem has multiple stages or cases
- You know conditional probabilities but not direct probability
- Keywords: "first..., then...", "depending on..."

Use Bayes when:

- You want to "reverse" a conditional probability
- You know $P(\text{effect}|\text{cause})$ but want $P(\text{cause}|\text{effect})$
- Keywords: "given that..., what caused it?"

Use indicator trick when:

- Computing $E[X]$ where X counts something
- Can break X into sum of events
- Especially good when events might be dependent

Use geometric distribution when:

- "Waiting time" until first success
- Repeated independent trials
- Keywords: "how many tries until...", "first time..."

8.2 Step-by-Step for Exam Problems

For probability problems:

1. Define events clearly (write out what A , B , etc. mean)
2. Identify if events are independent (check if one affects the other)
3. Look for partitions (use LOTP)
4. Check if you need to reverse conditional (use Bayes)

For expectation problems:

1. Can you write X as sum of indicators? (Do it!)

2. Is this a named distribution? (Use formula)
3. Can you condition on something simpler? (Use tower property)
4. Use linearity freely (no independence needed)

For variance problems:

1. First find $E[X]$
2. Then find $E[X^2]$ (often via sum of indicators squared)
3. Use $\text{Var}(X) = E[X^2] - (E[X])^2$
4. Check if variables are independent before adding variances!

9 Common Exam Mistakes to Avoid

1. **Thinking disjoint = independent.** NO! Disjoint events are dependent (if one happens, other can't).
2. **Forgetting linearity always works for $E[\cdot]$.** You can split expectations even when variables are dependent!
3. **Adding variances without independence.** $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ ONLY when independent.
4. **Not recognizing geometric distribution.** Keywords: "first success", "how many tries", "until..." $\Rightarrow$ probably Geometric.
5. **Forgetting memoryless property.** If problem says "already waited m trials" and asks about more waiting $\Rightarrow$ use memoryless: $P(X > m + n | X > m) = P(X > n) = (1 - p)^n$.
6. **Confusing $P(A|B)$ with $P(B|A)$.** These are different! To reverse, use Bayes.
7. **Not using indicator trick.** If you're counting events, write as sum of indicators: $X = \sum 1_{A_i}$, then $E[X] = \sum P(A_i)$.
8. **Forgetting $E[X^2] \neq (E[X])^2$.** These are only equal when $\text{Var}(X) = 0$.
9. **Computing expectations "the hard way".** Before computing $\sum x \cdot P(X = x)$, ask: Is this a named distribution? Can I use indicators? Can I condition on something?

10 Quick Reference Formulas

10.1 Must-Know Distributions

- Geom(p): $E = 1/p$, $P(X > k) = (1 - p)^k$, memoryless
- Bin(n, p): $E = np$, $\text{Var} = np(1 - p)$
- Pois(λ): $E = \lambda$, $\text{Var} = \lambda$
- Bernoulli(p): $E = p$, $\text{Var} = p(1 - p)$

10.2 Key Identities

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A^c) = 1 - P(A)$
- $P(A \cap B) = P(A|B)P(B)$
- $E[1_A] = P(A)$ (indicator trick)
- $\text{Var}(X) = E[X^2] - (E[X])^2$
- Independence: $X \perp Y \Rightarrow E[XY] = E[X]E[Y]$